

Section 6

Arousal, Sleep & Dreaming

Sexual Development

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COGS 17

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Arousal in the Brain

- **Reticular Formation**

- A set of interconnected nuclei that are located throughout the brainstem, synapses to various regions of the cortex
- Sensory information travels up through the Medulla & Pons and is distributed throughout the forebrain and cortex
- receives from all sensory systems
- Role: alerts brain, stimulates Thalamus and Basal Forebrain via ACh & Glutamate

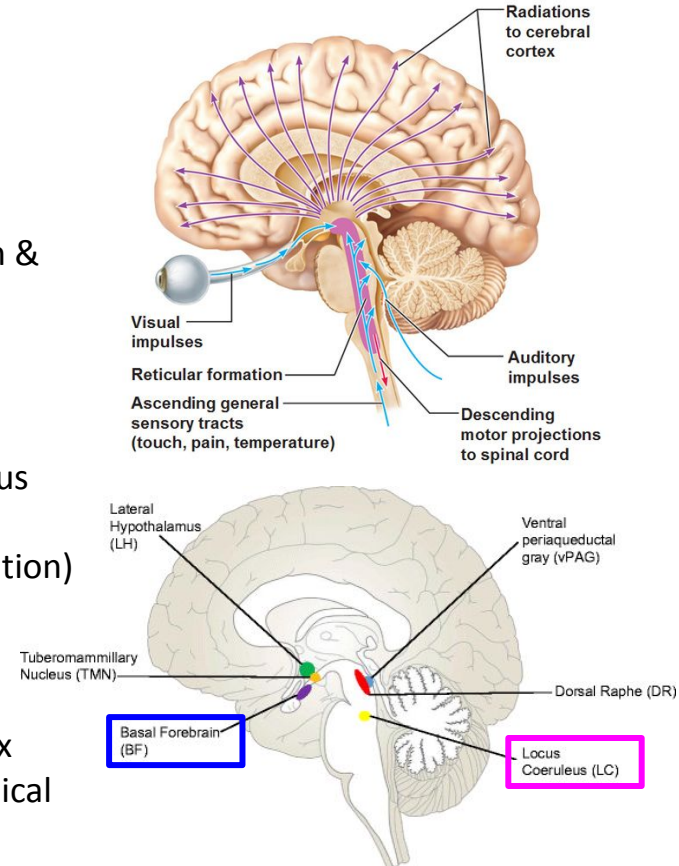
- **Locus Coeruleus**

- Part of the Reticular Formation in Pons
- widespread connections including Thalamus, Cortex & Hippocampus
- Releases bursts of Norepinephrine for vigilance and memory
- Active when learning new tasks & during vigilance (sustained attention)

- **Basal Forebrain**

- Anterior and dorsal to the Hypothalamus
- Releases **ACh** thruout cortex for **arousal** and **GABA** to **inhibit** cortex
- Damage: implicated in Alzheimer's Disease → arousal could be critical to memory activation

The Reticular Formation

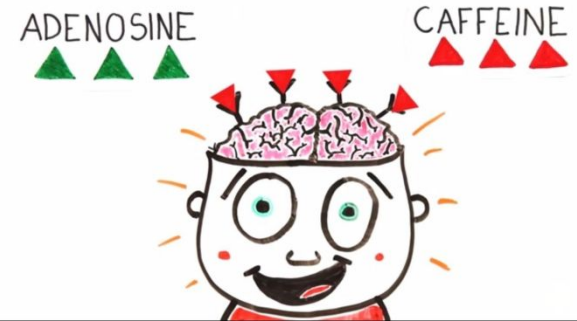
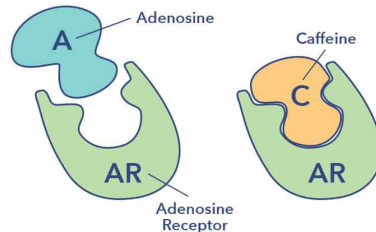


How does Caffeine help you wake up?



How does Caffeine help you wake up?

- Alertness/Sleepiness is controlled by an opponent system of ACh & GABA
- Basal forebrain controls the balance of these NTs
- Adenosine**
 - Its release inhibits Basal Forebrain's release of ACh, allowing GABA system to dominate
→ cortical activity suppression, promotes sleep
- Caffeine
 - crosses blood brain barrier
 - allows Basal Forebrain to continue to arouse the brain by blocking Adenosine Receptor sites
 - ACh sys. active → suppress GABA sys. → awake!



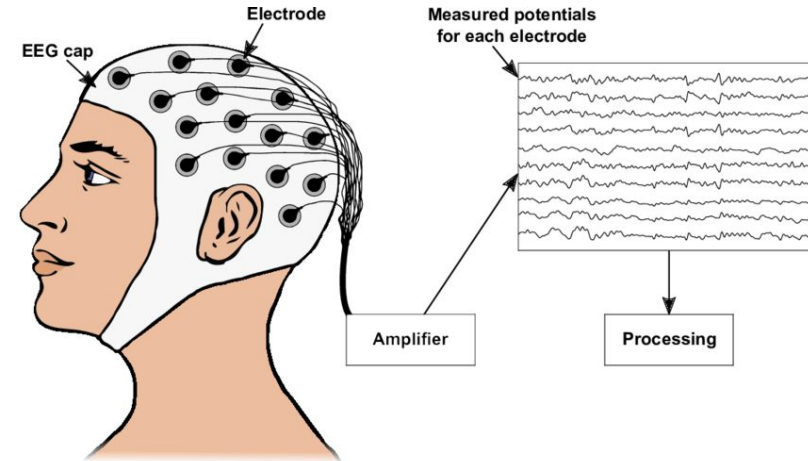
Caffeine only delays the onset of sleepiness, and only works before Adenosine binds to its receptor (i.e., when you're sleepy)

Stages of Wakefulness / Sleep

One way to measure brain activity during different states...

Electro-Encephalogram (EEG): gross average of electrical potentials in an area under electrode

- Electrodes are attached to the scalp
- Records
 - 1) Amplitude = Voltage
 - 2) Frequency = # of changes in average potential/time (Hz)
- high frequency, variable voltage waves → neural de-synchrony (multiple sources of stimulation, “many pebbles”)
- low frequency, high voltage waves → neural synchrony (coherent source of stimulation, “one big rock”)



https://www.researchgate.net/publication/338423585_Towards_a_home-use_BCI_fast_asynchronous_control_and_robust_non-control_state_detection?_tp=eyJjb250ZXh0Ijp7ImZpcnN0UGFnZSI6Ii9kaXJlY3QILCJwYVWdlIjoieXZlRmVjdCJ9fQ

Stages of Wakefulness / Sleep

- Awake, active: **Beta** waves (18-24 Hz)
- Awake, relaxed: **Alpha** waves (8-12 Hz), “psychological clarity” after meditation, best performance (reaction time, memory...)
- **Sleep Cycle**: Multiple stages
 - Stage 1 Sleep: **Theta** waves (4-7 Hz)
 - Stage 2 Sleep: Theta waves + **Spindles** (rapid changes) and **K Complexes** (slow wave)
 - Stages 3 & 4: **Delta** waves (<4 Hz), Slow Wave Sleep (SWS)
 - Stage 3 if “Slow wave sleep (= < 4Hz)” = < 50%, very synchronized
 - Stage 4 if = > 50%, hardest to wake
 - As more sleep cycles progress, Stage 3 and 4 durations decrease
- Rapid Eye Movement (REM) Sleep: Variable EEG (12-28 Hz)

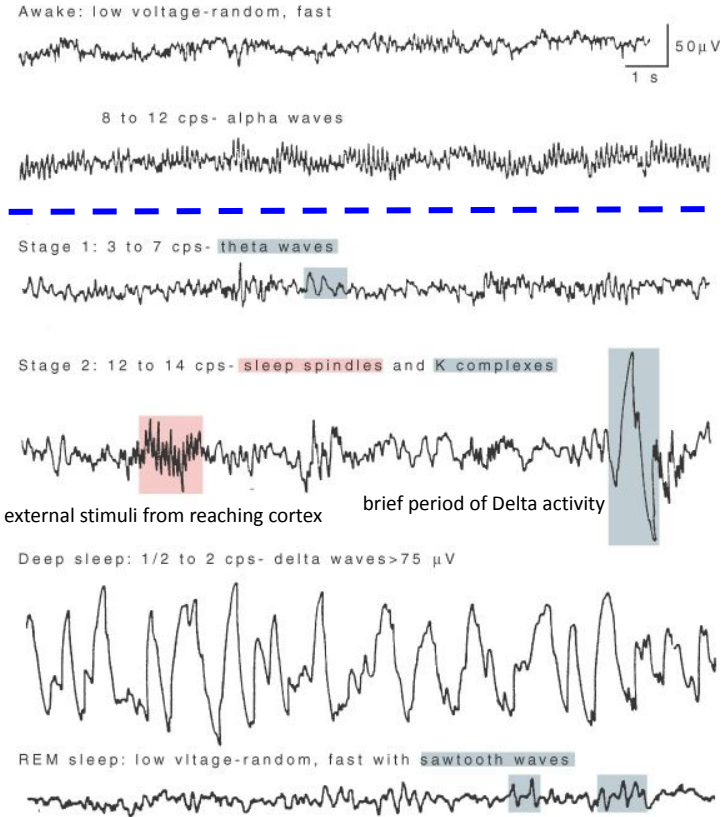


FIGURE 42.4 Electroencephalograms showing electrical activity of the human brain during different stages of sleep.

Stages of Wakefulness / Sleep

desynchronized
(lob many stones into a pond)

so as move into deeper sleep...

frequency decreases,
voltage increases
as brain activity becomes synchronized

synchronized
(lob one stone into a pond)

most desynchronized
(lob imaginary stones into a pond)

Awake: low voltage-random, fast



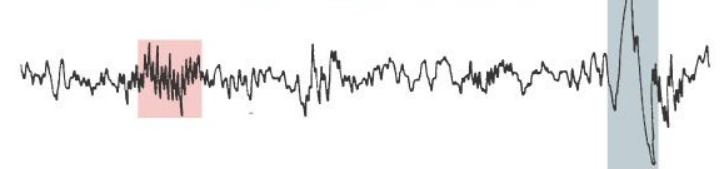
8 to 12 cps- alpha waves



Stage 1: 3 to 7 cps- theta waves



Stage 2: 12 to 14 cps- sleep spindles and K complexes



Deep sleep: 1/2 to 2 cps- delta waves >75 μV



REM sleep: low voltage-random, fast with sawtooth waves

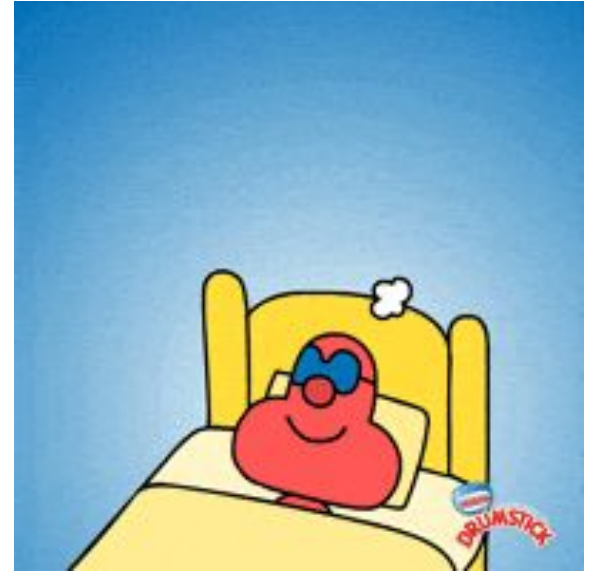


FIGURE 42.4 Electroencephalograms showing electrical activity of the human brain during different stages of sleep.

Rapid Eye Movement (REM)

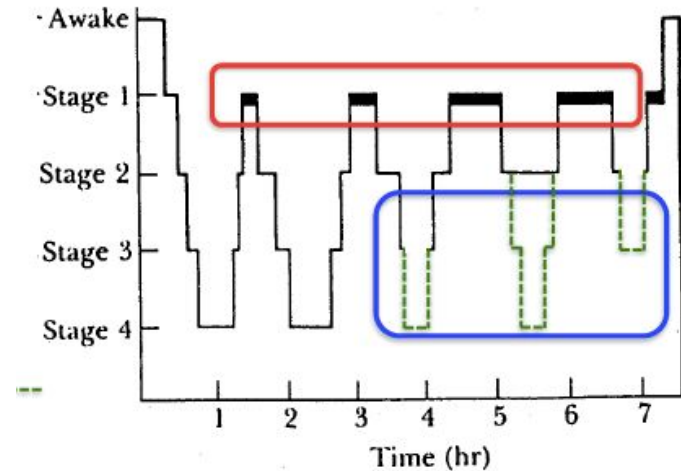
a.k.a Paradoxical Sleep, because of contradictory set of conditions that occur:

- EEG is desynchronized: high frequency (like Sleep 1 or even Awake), low voltage waves (only Sleep 1 is lower)
- heart/breathing rate, blood pressure is more variable than in other sleep stages
- eyes move, genitals active, but muscles are paralyzed (**Atonia**)
- may have cognitive/psychological advantages (e.g., REM-deprived → poorer memory for previous day's learning!)
- highly correlated with dreaming (but not 100%)



Changes in Sleep Cycles

- ~ 90 minute cycle from stage 1 to REM (Stage 1,2,3,4,3,2,REM)
- As night progress, duration of Stage 3&4 decreases, whereas REM duration increases (dreaming)
- Sleep Deprivation → more slow wave sleep; Lethargy, Poor Concentration, Irritability, Memory impairment, Decreased resistance to infection
- REM Deprivation
 - System attempts to enter REM more frequently and bypasses early stages of sleep (REM rebound)
 - Continued → psychosis, hallucinations, death
- Functions of Sleep & Dreaming are still largely not well defined
 - but... Sleep is restorative (engage in critical metabolic processes, process waste, etc)
 - REM might warm sleeping brain and Dreaming might help consolidate memories and resolve psychological conflicts

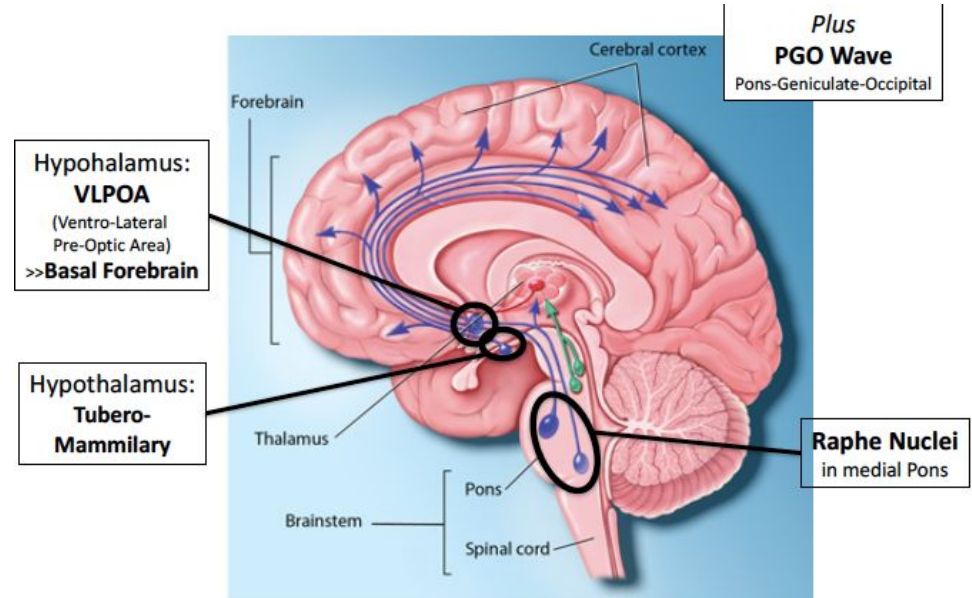


Dream is a Mystery



Neural Control of Sleep

- **Hypothalamus** nuclei are critical in initiating and regulating sleep!
 1. **Ventrolateral Preoptic Area (VLPOA)** releases **GABA** to inhibit the brainstem via the basal forebrain and the cortex
 1. **Tubero-Mammillary body** releases excitatory Histamine 1 after periods of slow wave sleep, indirectly enables initiation of ACh release from Pons >> initiates PGO wave



Neural Control of Sleep

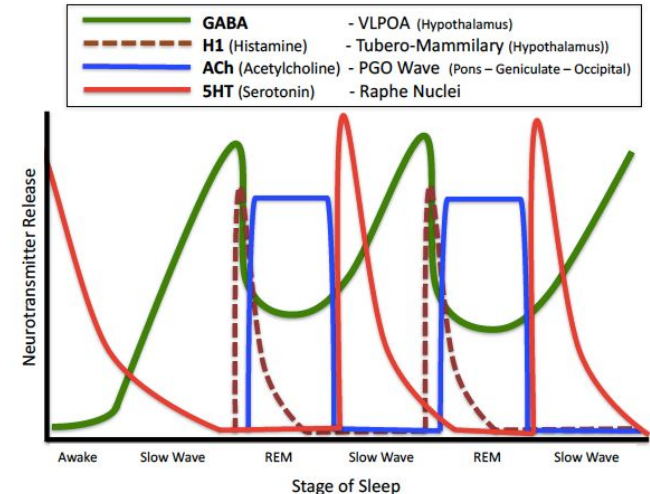
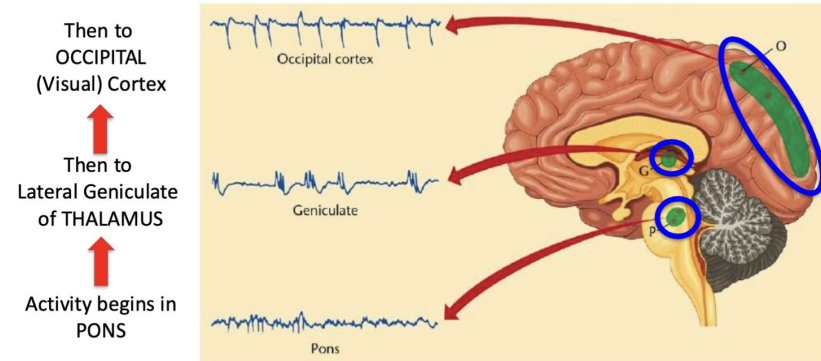
- **Pons, Geniculate, Occipital (PGO) Wave**

- A sequence of activity in **Pons** → Lateral **Geniculate** Nucleus of Thalamus → **Occipital Cortex** - **initiates REM**
- how does this sys. work: excitatory ACh arouses and desynchronizes visual & other sensory/motor pathways
- ACh builds up just before REM, holds steady, then falls back to lower levels

- **Raphe Nuclei in medial Pons**

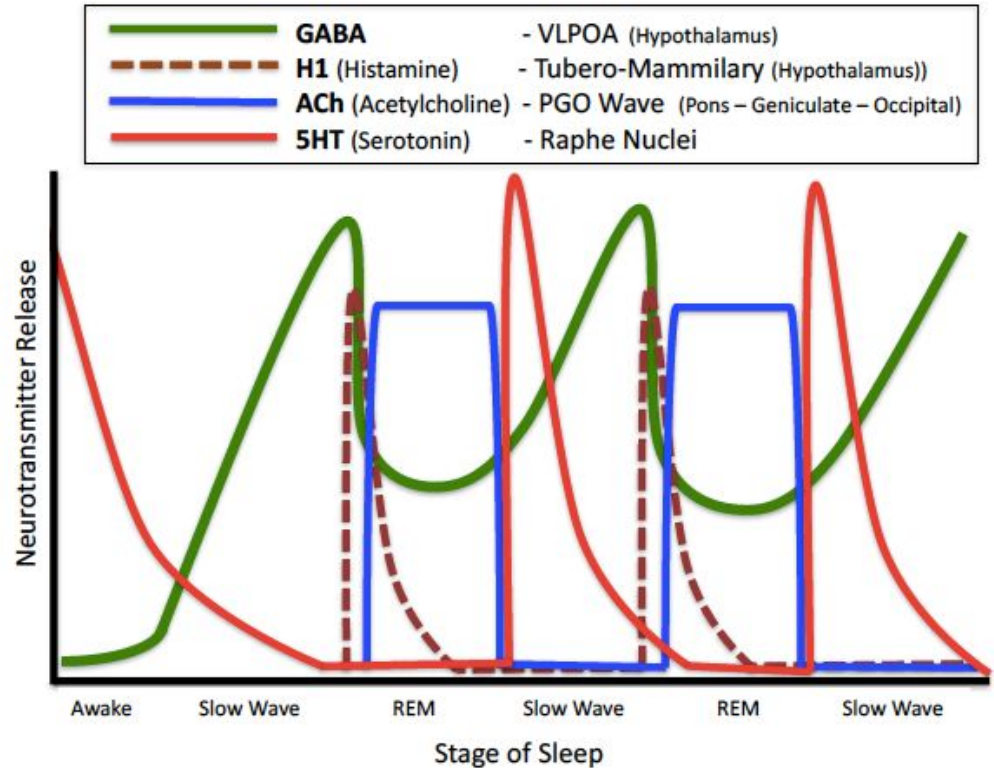
- Produces serotonin (5HT)
- Decrease in serotonin output → sleepiness
- In sleep cycle: very low serotonin during sleep, none during REM, but sudden burst that shuts REM off
- 5HT gradually falls, shifting system back to slow wave sleep
- when 5HT flat, PGO wave is initiated again by Pons via ACh

>> Stages of sleep are controlled by an interaction of different types & sources of neural activity



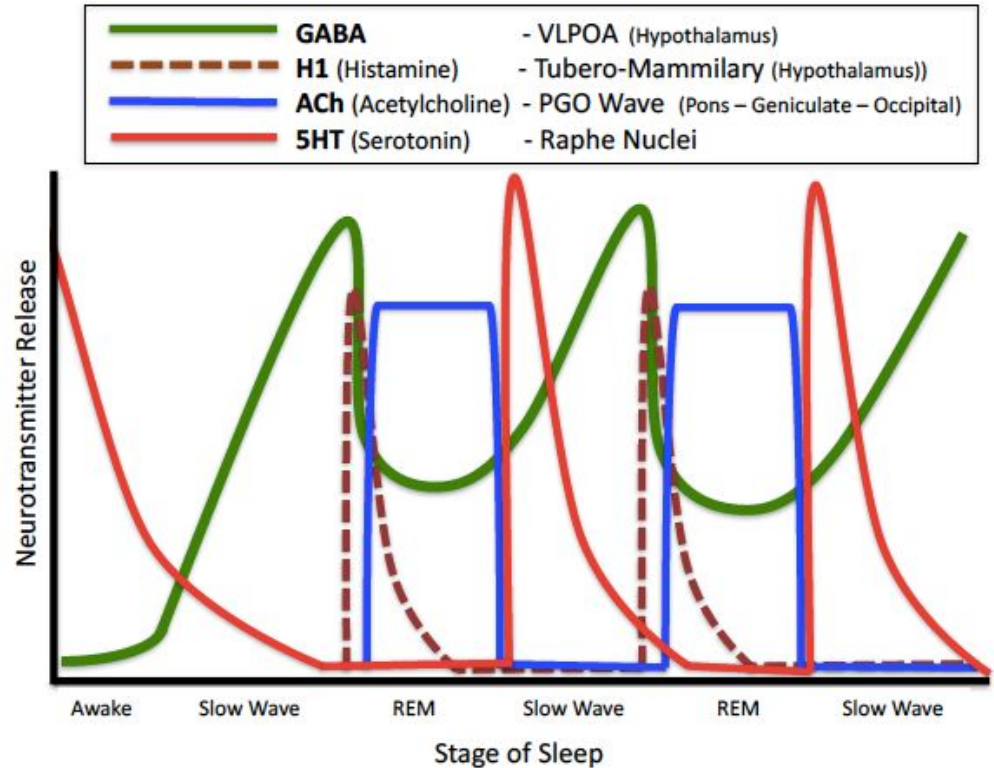
Neurotransmitters in Sleep

- **GABA**
 - Initiated by Ventrolateral Preoptic Area
 - Climbs during slow wave, low in REM
- **Histamine (H1)**
 - Tubero-Mammillary body in Hypothalamus
 - Spikes for the brain to warm up and initiate REM
- **ACh (Acetylcholine)**
 - High during REM
- **5HT (Serotonin)**
 - once warmed up, blast of 5HT in Raphe Nuclei shuts off PGO wave (“turn off” REM)



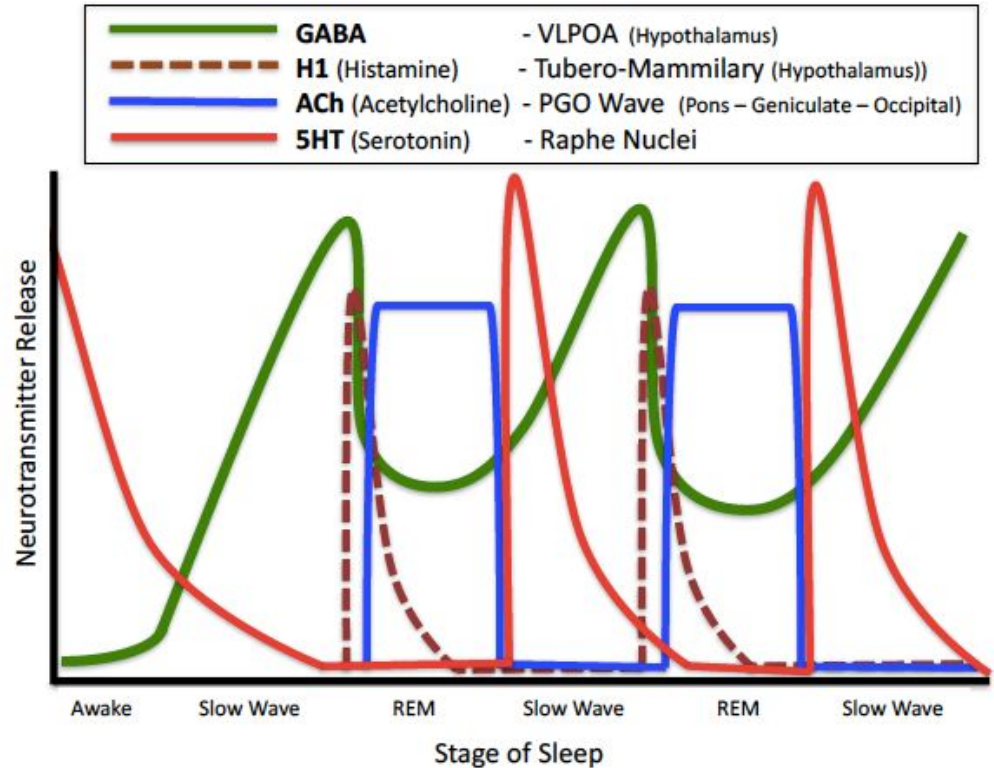
Neurotransmitters in Sleep

1. _____ arouses cortex during dreams
1. _____ promotes sleep
1. Burst of _____ shuts off REM
1. _____ is released when brain cools



Neurotransmitters in Sleep

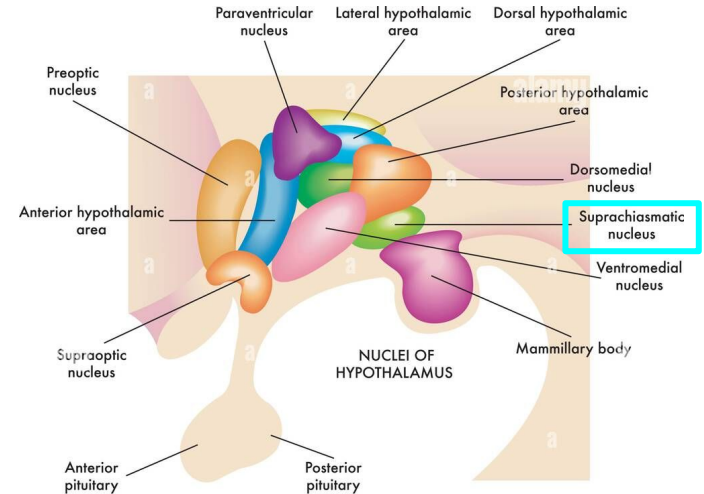
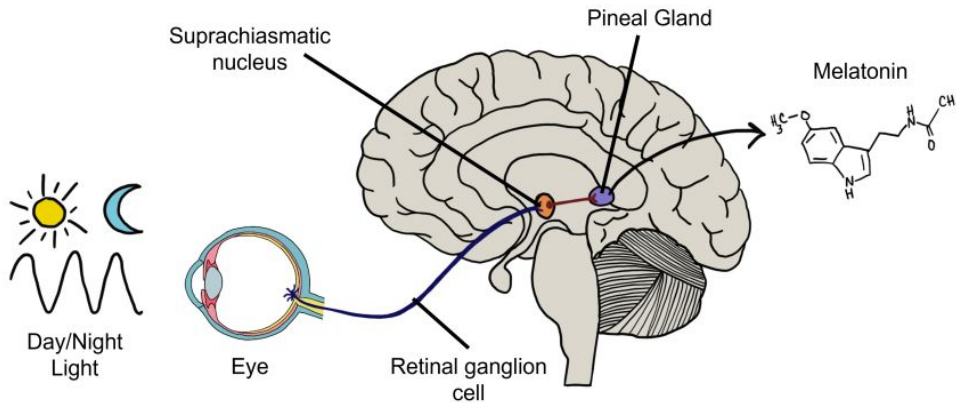
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1. **Histamine** is released when brain cools



Biological Rhythms

: what regulates sleep and arousal sys, mediated by interaction b/w **Suprachiasmatic Nucleus & Pineal Gland**

- Suprachiasmatic Nucleus (SCN)** of Hypothalamus (just anterior to Optic Chiasm) = Circadian clock
 - Establishes rhythm of $\sim 24 \pm 1$ hours
 - very robust: even if blinded, food/water/oxygen-deprived
 - Releases hormones into bloodstream & projects to other Hypothalamic Nuclei, Brainstem, Pituitary glands
 - Cycle is regulated through projections to and from Pineal Gland

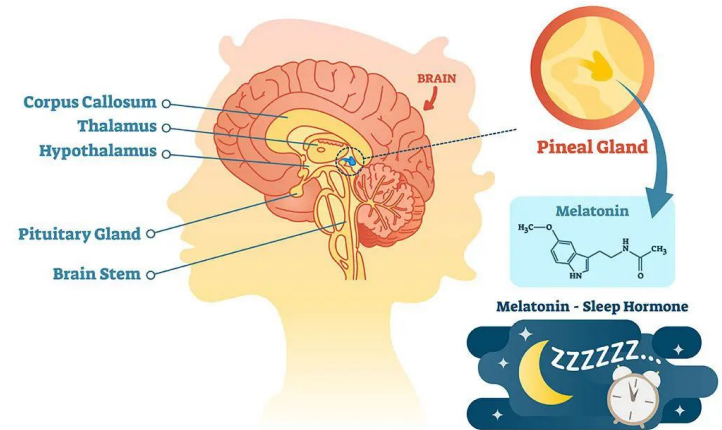
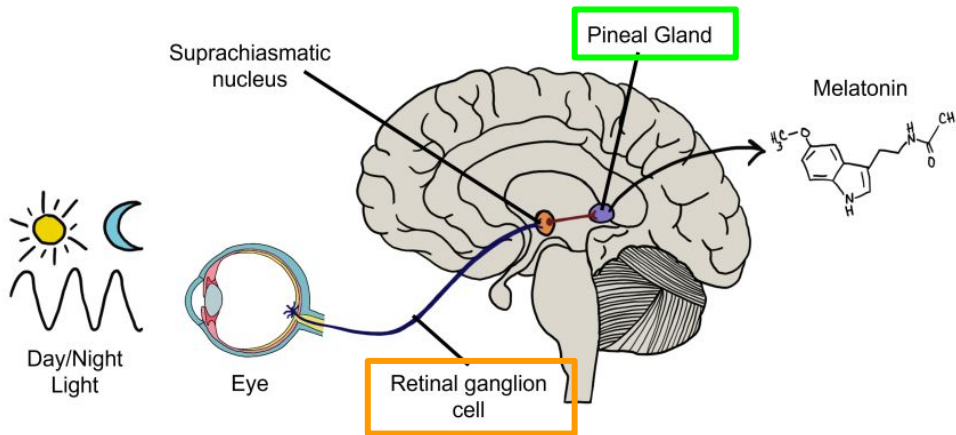


Biological Rhythms

: what regulates sleep and arousal sys, mediated by interaction b/w **Suprachiasmatic Nucleus & Pineal Gland**

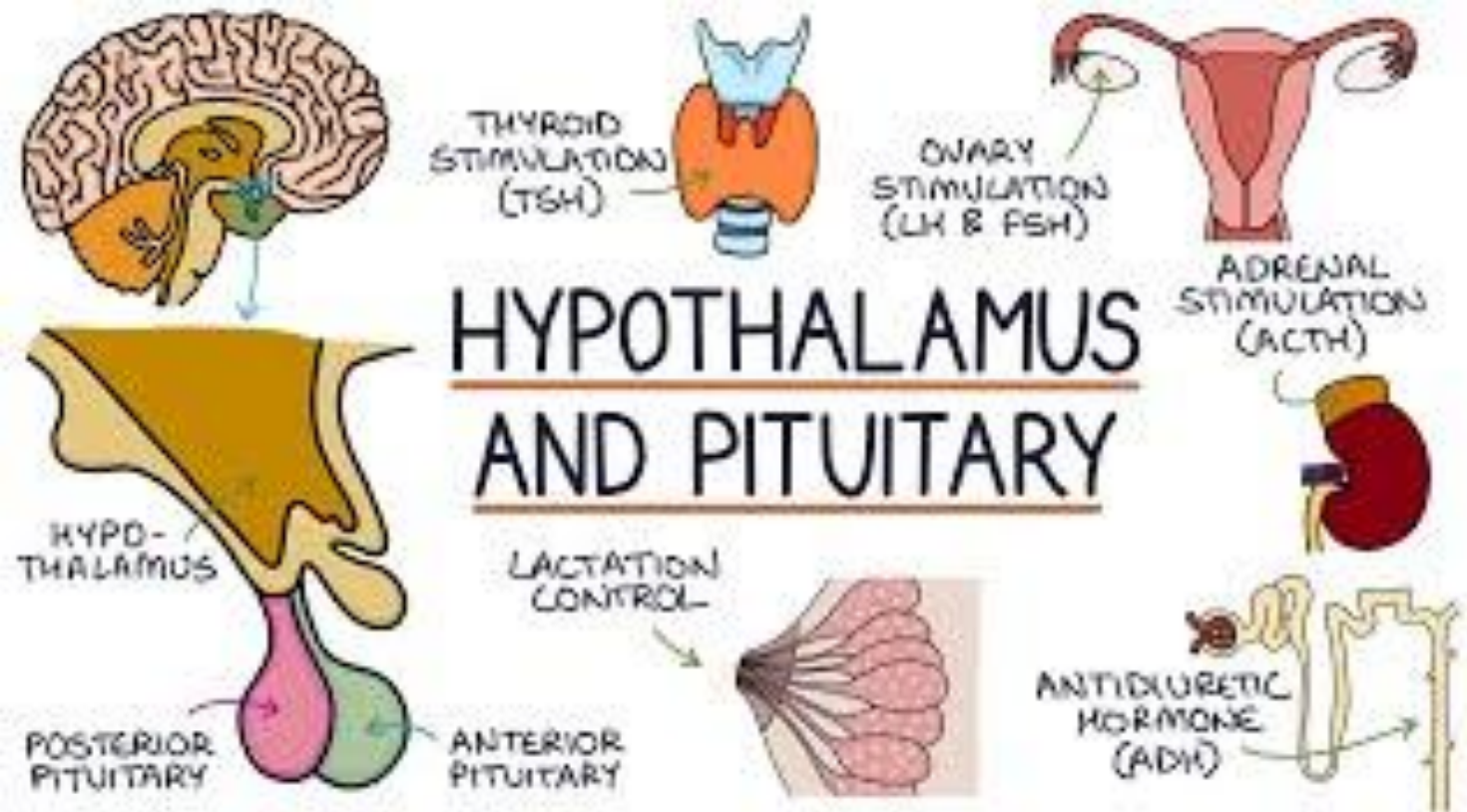
2. Pineal Gland (just superior to midbrain, posterior to Thalamus)

- decide “whether to produce hormone Melatonin or not?” which increases sleepiness
- SCN has receptor sites for Melatonin, so when Pineal increases Melatonin output at end of day → helps regulate cycle
- When light comes up in the morning, the light goes to SCN via Retino-Hypothalamic Path produces inhibitory output to Pineal → **decreases Melatonin production, allows wakefulness**



Sexual Development & Behavior

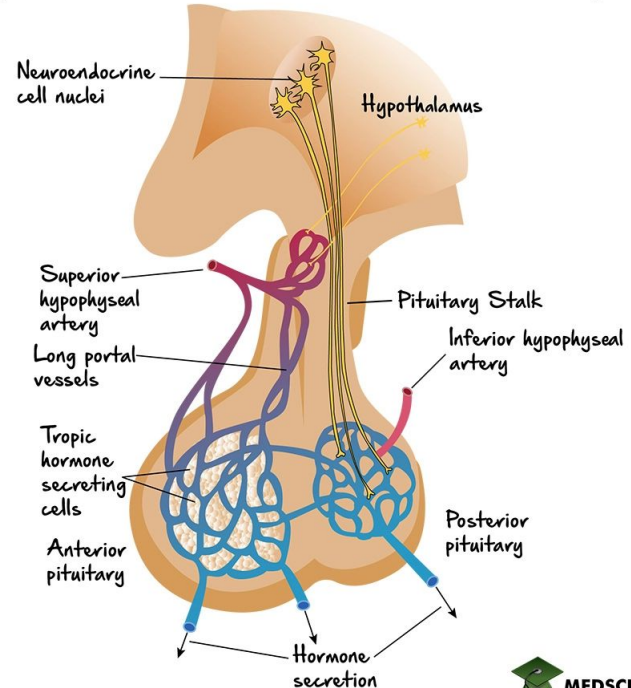
Hypothalamus



Hypothalamus

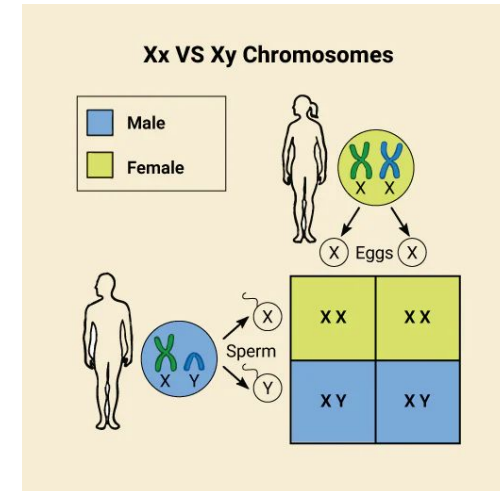
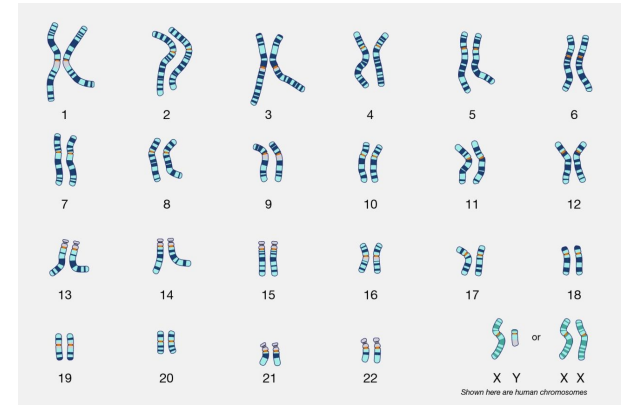
- Anatomically and functionally related to the **pituitary gland**
- controls endocrine systems via commanding Pituitary Gland
- produces **Releasing Hormones** that flow via blood vessels to Anterior Pituitary > “Hey Pituitary, RELEASE hormones”
- Releases other hormones as NTs via axons to posterior pituitary
- Reproductive Hormones have Organizing effects (on anatomy in fetal development & puberty) & Activating effects (influence behavior)
- Communicates to adrenal gland, and gonads (ovaries/testis)

Hypothalamus and the Pituitary Gland



Chromosomes

- 23 pairs inherited from your parents
- Chromosomes are necessary but insufficient to determine gender
- Female: XX, Male XY, but hormone activity is required to determine gender!
- **Steroid hormones**
 - **Androgens**: General term to describe **male** hormones (e.g., testosterone)
 - **Estrogens**: General term to describe **female** hormones (e.g., estradiol)
- But we all have both kinds! (mix of Androgens and Estrogens in varying proportions)
 - Males have more androgens than estrogens
 - Females have more estrogens than androgens



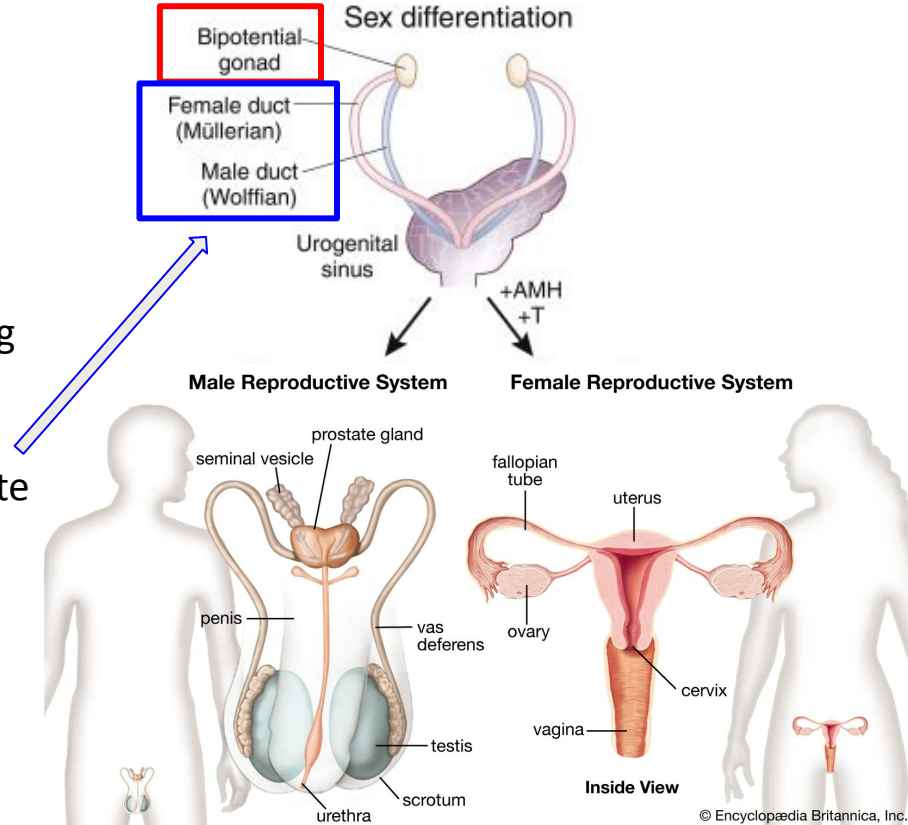
Organizing Effects

Fetal Development of Sexual Anatomy

- every mammalian fetus has the anatomical precursors for both sexes

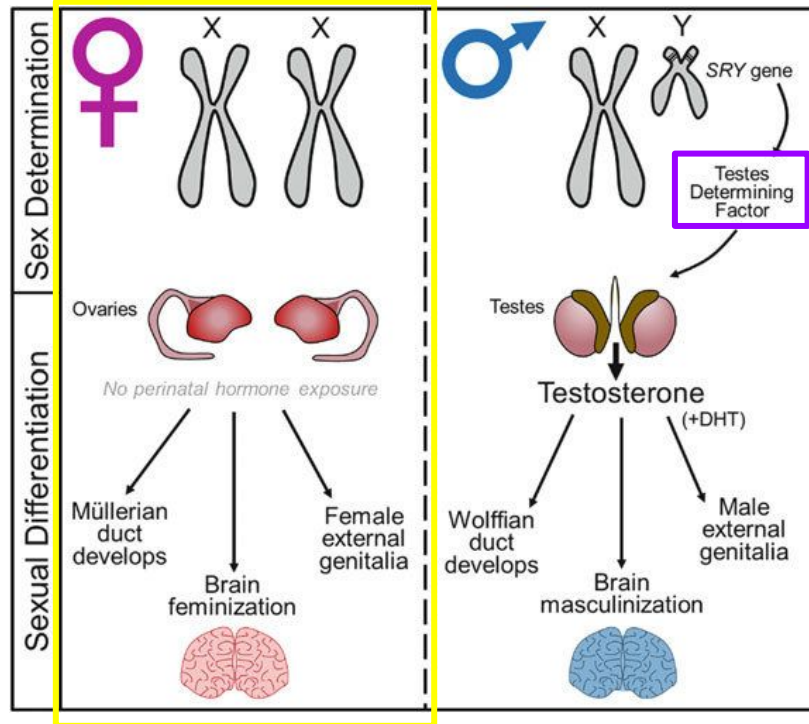
e.g., same initial structure...

1. **Gonads** (later become either Testes or Ovaries)
2. **Gentalia** (later become M/F anatomy, depending on Androgens/Not)
3. Each fetus has both
 - a. (M) Wolffian duct → Vas Deferens & Prostate
 - b. (F) Muellerian duct → Fallopian Tubes & Uterus
 - c. other system degenerates and these ducts development also depends on Androgens/Not



Organizing Effects

- The genes controlling M/F body & brain development are also present in both sexes, except...



the "SWITCH"

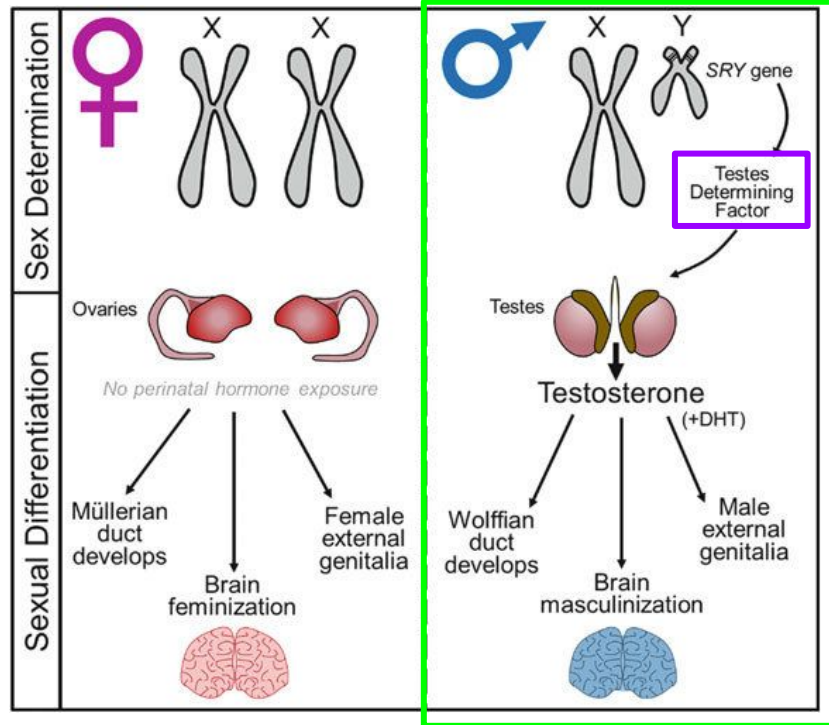
- found only on Y chromosome
- signals production of the **Testis-Determining Factor (TDF) Enzyme**
- occurs during Critical Period of fetal development (TDF appears 6-8th week, genitals developed by 4th month)

If TDF is **NOT** present → Female

- ovaries differentiate, Müllerian sys develops, Wolffian regresses, female genitalia develop
- But what if...
 - fetus is XY, but lacks specific gene for TDF
 - fetus is XO (Turner's Syndrome, no Y chromosome): both will develop internally & externally as female, but will be infertile, since two Xs are required to produce ova (eggs)

Organizing Effects

- The genes controlling M/F body & brain development are also present in both sexes, except...



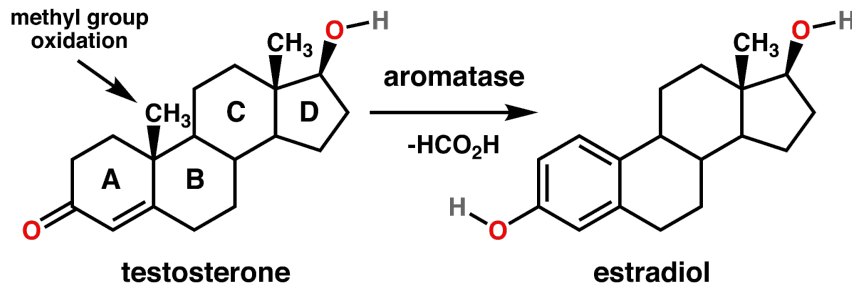
If TDF is present → Male

- testes differentiate, producing Androgens, Wolffian ducts and male genitalia develop
 - testes also produce Anti-Mullerian Hormone (AMH -inhibits Mullerian sys development)
- If XY fetus is Androgen-insensitive, testes still produce Androgens and AMH
 - but the lack of androgen receptors inhibits Wolffian system development
 - while AMH is still functioning → inhibit the Mullerian system

→ no internal sex organs and so will be infertile
- If XX fetus is exposed to Testosterone during critical period, develops (semi-)male form, sometimes infertile
 - some tendency for these individuals to be homosexual (+ body, brain, societal influences)?

Estrogen and its Masculinizing Effect

- When testosterone enters a fetal cell, it is “aromatized” or converted into Estrogen, which leads to male development
- One can ask... if the Mother has Estradiol present, why aren't all offspring masculinized?
 - If there is exposure to estradiol from the mother, fetus' **Alpha-Fetoprotein** (AFP) protein binds with mothers' estrogens, preventing it from influencing the fetus's sexual development
 - AFP: produced by the fetal liver, allows the fetus's own hormones to determine its gender
- But... when excessive estrogens administered (e.g., DES, synthetic estrogen that was prescribed to pregnant women between 1938 and 1971 to prevent miscarriage), this safeguarding protective sys can be overwhelmed → potential masculinization in the fetus



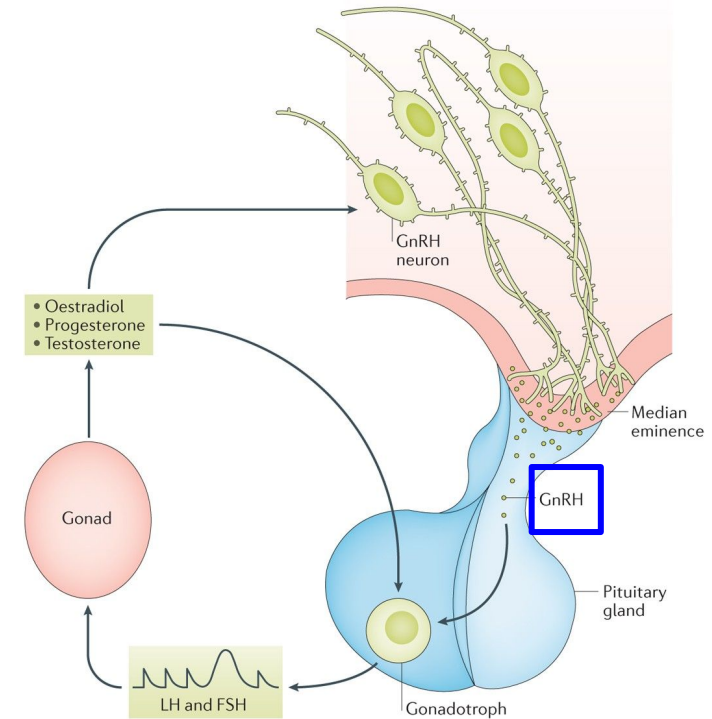
Testosterone and Estrogen are VERY similar

In fact, Testosterone can be converted into estrogen (thru an enzyme called **aromatase**)

Secondary Sexual Characteristics

During Adolescence

- In both sexes, hypothalamus releases **Gonadotrophin-Releasing Hormones (GnRH)**
 - causing Anterior Pituitary to release Gonadotropic Hormones, LH and FSH (Leutinizing & Follicle Stimulating Hormone)
 - Present in both males and females
- LH and FSH stimulate the testes in **males** to produce sperm and testosterone (and other androgens, and low levels of estrogens)
→ adult male form
- LH and FSH stimulate the ovaries in **females** to release ova and estradiol (and other estrogens, and low levels of androgens)
→ adult female form
 - In females, Androstenedione (androgen) is released by adrenal glands for secondary hair growth

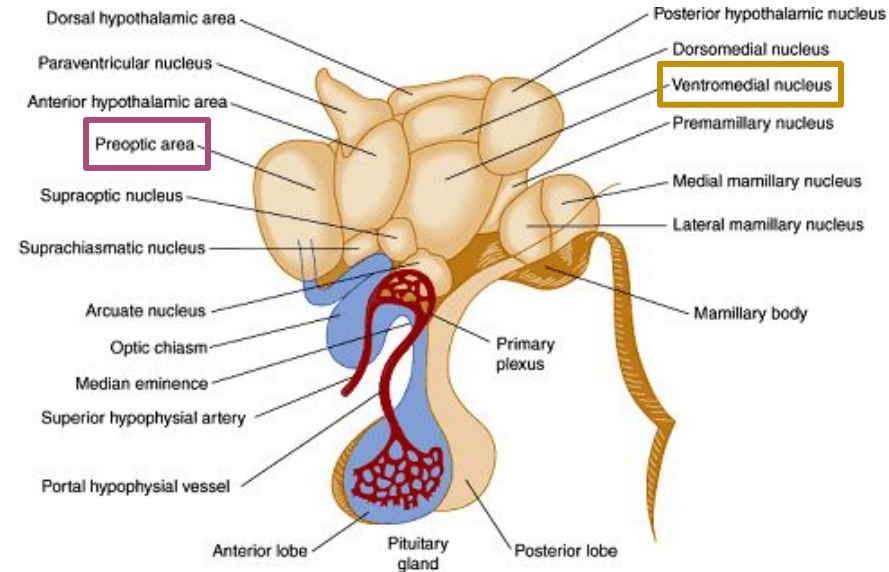


Sexual Differences in Brain Development

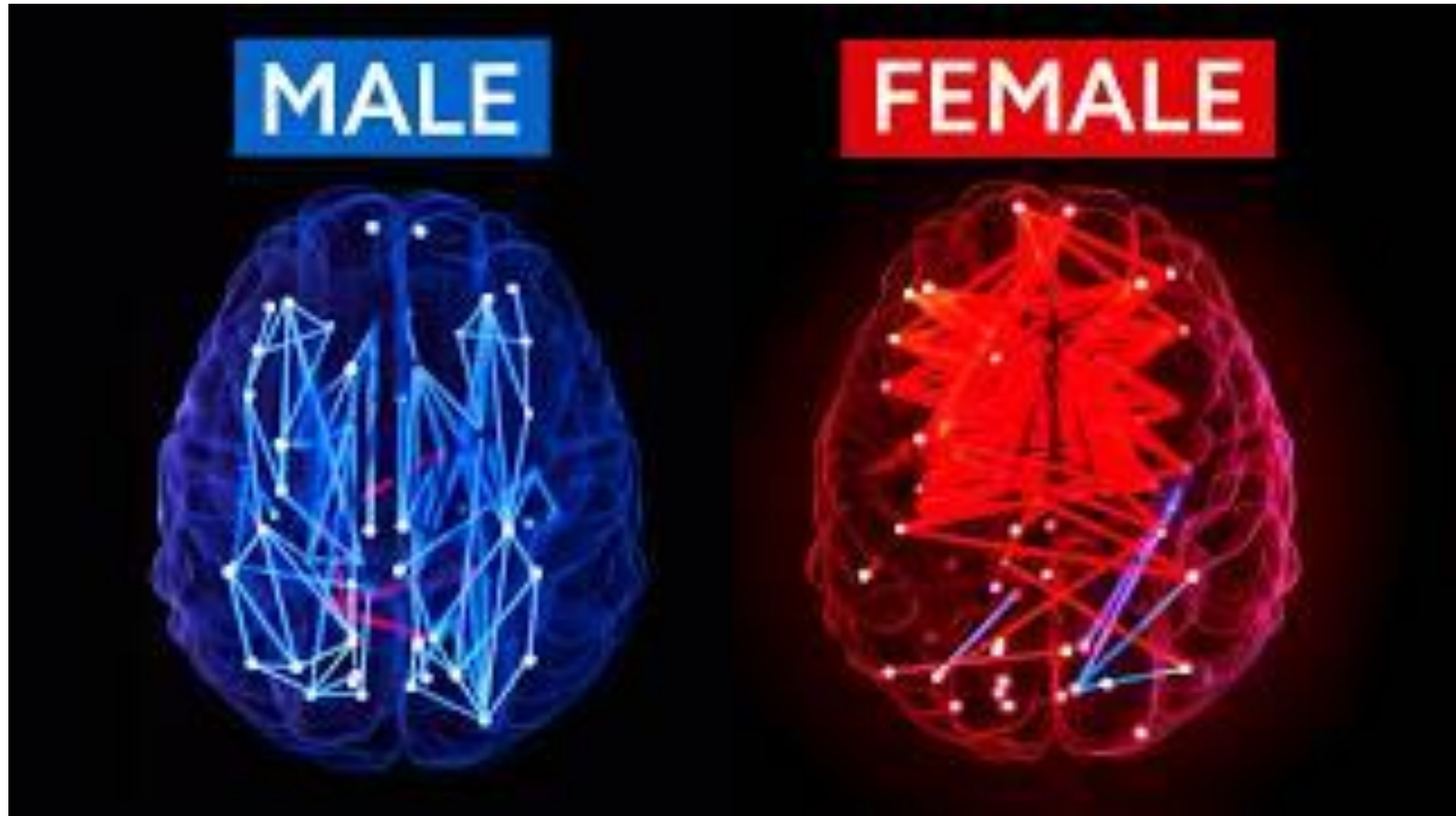
- Presence/absence of testosterone during prenatal period and early infancy → differences in brain

Hypothalamus

- **Medial Preoptic Area (MPOA)**
 - Has androgen receptor sites, esp. active during Male sexual behavior
 - Includes the **Sexually Dimorphic Nucleus (SDN)**, 2.5X larger in males (vs. females)
 - Early testosterone is required for SDN development and without it, males will not respond to androgen activity
- **Ventromedial Nucleus (VMH)** is larger in females than males
 - Has estrogen receptor sites, esp. active during Female sexual behavior
 - Develops in absence of early Testosterone
 - Also regulates feeding behavior or control of eating



Gender differences in Brains



Happy Thanksgiving!

